Chinmay's watchdog + reset bricks — one sheet per supervisor IC

Each sheet replicates 'WD and RST TPS3823': supervisor -> 74LVC1G32 (PROG bypass) -> 74LVC1G07 open drain -> nRST. Sheet pins: WDI (in), nRST (out).



WD and RST TPS3823

WDI

nRST

TPS3823-33DBVR

Texas Instruments

File: WD_RST_TPS3823.kicad_sch

WD and RST TPS3828

WDI

nRST

TPS3828-33DBVR

Texas Instruments

File: WD_RST_TPS3828.kicad_sch

WD and RST TPS3824

WDI

nRST

TPS3824-33DBVR

Texas Instruments

File: WD_RST_TPS3824.kicad_sch

WD and RST STM6822

WDI

nRST

STM6822TWY6F

STMicroelectronics

File: WD_RST_STM6822.kicad_sch

WD and RST STM6823

WDI

nRST

STM6823TWY6F

STMicroelectronics

File: WD_RST_STM6823.kicad_sch

WD and RST STM6824

WDI

nRST

STM6824TWY6F

STMicroelectronics

File: WD_RST_STM6824.kicad_sch

WD and RST MCP1316

WDI

nRST

MCP1316T-29LE/OT

Microchip

File: WD_RST_MCP1316.kicad_sch

WD and RST MCP1316M

WDI

nRST

MCP1316MT-29LE/OT

Microchip

File: WD_RST_MCP1316M.kicad_sch

WD and RST MCP1318

WDI

nRST

MCP1318T-29LE/OT

Microchip

File: WD_RST_MCP1318.kicad_sch

WD and RST MCP1320

WDI

nRST

MCP1320T-29LE/OT

Microchip

File: WD_RST_MCP1320.kicad_sch

WD and RST MCP1321

WDI

nRST

MCP1321T-29LE/OT

Microchip

File: WD_RST_MCP1321.kicad_sch

WD and RST TPS35

WDI

nRST

TPS35CA40KAGDDFR

Texas Instruments

File: WD_RST_TPS35.kicad_sch

WD and RST TPS3851

WDI

nRST

TPS3851G33SDRBR

Texas Instruments

File: WD_RST_TPS3851.kicad_sch

WD and RST TPS3850

WDI

nRST

TPS3850G33DRCR

Texas Instruments

File: WD_RST_TPS3850.kicad_sch

WD and RST MAX16056

WDI

nRST

MAX16056ATA31+T

Analog Devices

File: WD_RST_MAX16056.kicad_sch

WD and RST ADM8613

WDI

nRST

ADM8613Y232ACBZ-R7

Analog Devices

File: WD_RST_ADM8613.kicad_sch

WD and RST MAX6369

WDI

nRST

MAX6369KA+T

Analog Devices

File: WD_RST_MAX6369.kicad_sch

APPROVED PASSIVES (reused across all sheets: same value + rating + operating voltage = same MPN)
CAP-0603-100N-50V-X7R x51 CL10B104KB8NNNC (Samsung Electro-Mechanics) | alt: C0603C104K5RACTU (KEMET), GCJ188R71H104KA12D (Murata), 885012206095 (Wurth Elektronik)
CAP-0603-39N-50V-X7R x1 CL10B393KB8NNNC (Samsung Electro-Mechanics) | alt: CC0603KRX7R9BB393 (Yageo), 06035C393KAT2A (KYOCERA AVX), GCJ188R71H393KA12D (Murata)
CAP-0603-330N-16V-X7R x1 CL10B334KO8NNNC (Samsung Electro-Mechanics) | alt: C1608X7R1C334K080AC (TDK), C0603C334K4RACTU (KEMET), AC0603KRX7R7BB334 (Yageo)
CAP-0805-10U-25V-X5R x1 GRM21BR61E106KA73L (Murata) | alt: CL21A106KAYNNNE (Samsung Electro-Mechanics), C2012X5R1E106K085AC (TDK), CC0805KRX5R8BB106 (Yageo)
RES-0603-10K-5%-0.1W x27 RC0603JR-0710KL (Yageo) | alt: ERJ-3GEYJ103V (Panasonic), CRCW060310K0JNEA (Vishay Dale), CR0603-JW-103ELF (Bourns)
Derating rules: C applied V <= 50% of rated, dielectric Tmax >= Tamb+20C; R power <= 50% of derated rating at 85C, V <= 80% of max working V
C1 (X5R, 85C max) FAILS the temperature rule at Tamb 85C: test-bench only, not part of any block

Replicates block: WD and RST TPS3823

Chinmay

Sheet: /

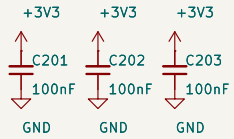
File: Chinmay_WD_RST.kicad_sch

Title: Chinmay WD and RST library

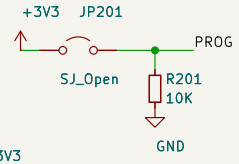
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 1/18

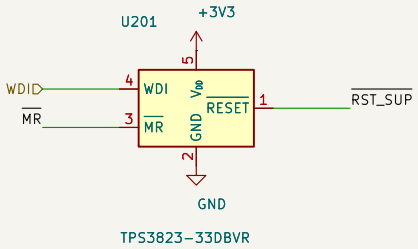
Watchdog and Reset controller – TPS3823-33DBVR

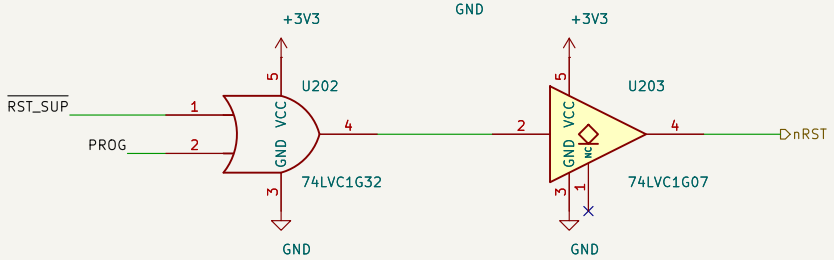
Decoupling: one 100nF per IC

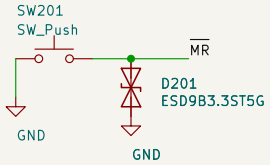


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C201 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C202 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C203 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R201 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

TPS3823 supervisor, 2.93 V threshold, 1.6 s watchdog, push-pull $\overline{\text{RESET}}$, manual reset
Group 1 SOT-23-5 (MAX823 pinout). Drop-in: STM6823, MCP1316.
Push button drives MR
Alternates (MPN2): STM6823TWY6F, MCP1316T-29LE/OT

Replicates block: WD and RST TPS3823

Chinmay

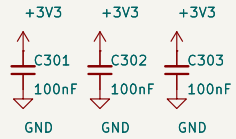
Sheet: /WD and RST TPS3823/
File: WD_RST_TPS3823.kicad_sch

Title: WD and RST TPS3823

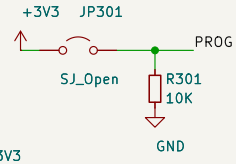
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 2/18

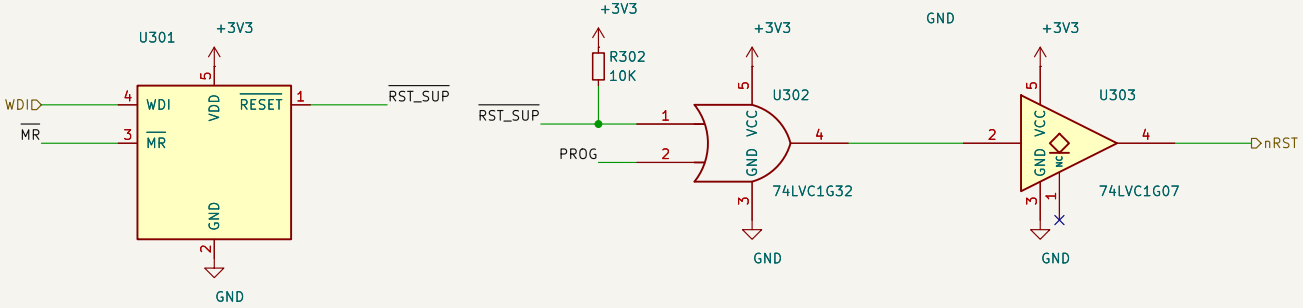
Watchdog and Reset controller – TPS3828-33DBVR

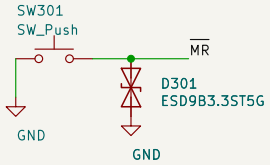
Decoupling: one 100nF per IC



Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash







PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)

C301	100nF	CL10B104KB8NNNC	V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C302	100nF	CL10B104KB8NNNC	V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C303	100nF	CL10B104KB8NNNC	V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R302	10K	RC0603JR-0710KL	P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R301	10K	RC0603JR-0710KL	P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK

Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

TPS3828 supervisor, 2.93 V threshold, 1.6 s watchdog, open-drain $\overline{\text{RESET}}$, manual reset
Group 1 SOT-23-5_open-drain output: R pull-up fitted.
Push button drives MR
Alternates (MPN2): STM6822TWY6F, MCP1320T-29LE/OT

Replicates block: WD and RST TPS3823

Chinmay

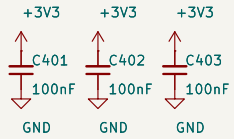
Sheet: /WD and RST TPS3828/
File: WD_RST_TPS3828.kicad_sch

Title: WD and RST TPS3828

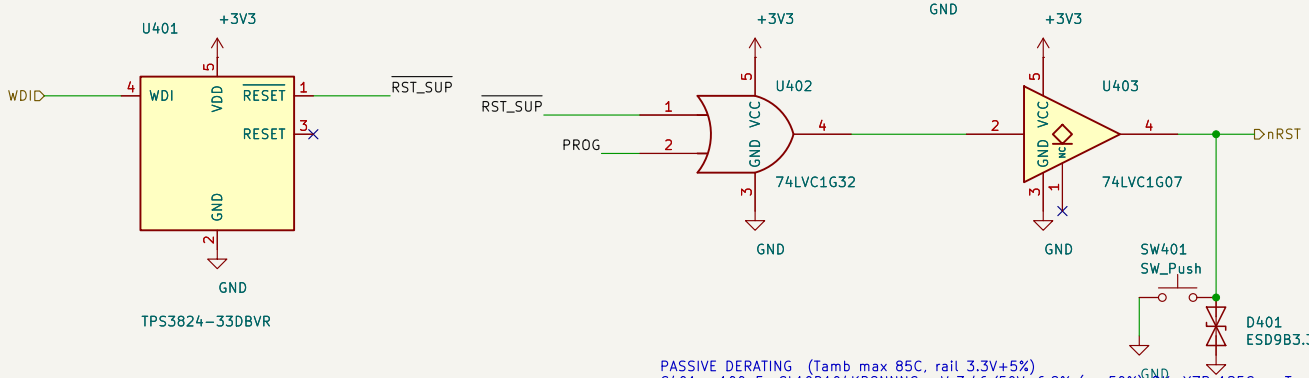
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 3/18

Watchdog and Reset controller — TPS3824-33DBVR

Decoupling: one 100nF per IC



Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash



PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C401 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C402 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C403 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R401 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

TPS3824 supervisor, 2.93 V threshold, 1.6 s watchdog, push-pull $\overline{\text{RESET}}$ + RESET, no MR
Group 2 SOT-23-5 (MAX824 pinout): no MR pin, button resets nRST directly.
No MR pin on this IC: push button pulls nRST low directly
Alternates (MPN2): STM6824TWY6F, MCP1318T-29LE/OT

Replicates block: WD and RST TPS3823

Chinmay

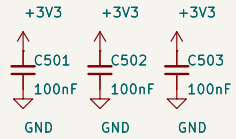
Sheet: /WD and RST TPS3824/
File: WD_RST_TPS3824.kicad_sch

Title: WD and RST TPS3824

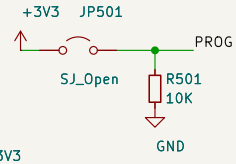
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 4/18

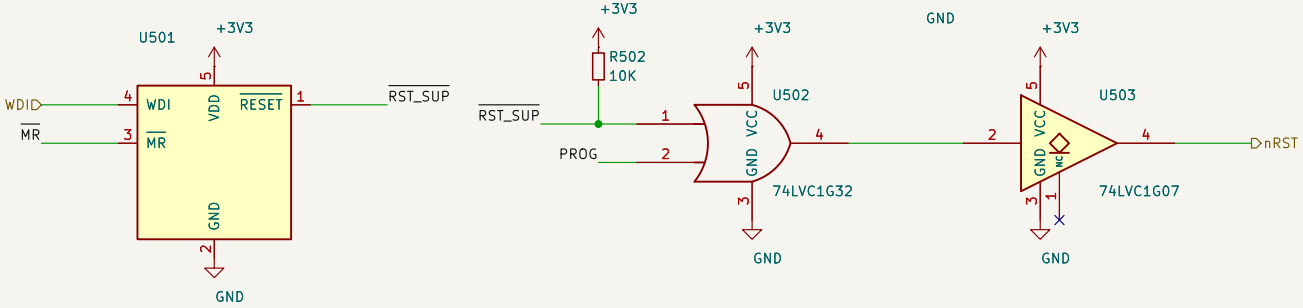
Watchdog and Reset controller — STM6822TWY6F

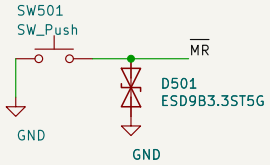
Decoupling: one 100nF per IC



Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash







PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)

C501	100nF	CL10B104KB8NNNC	V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C502	100nF	CL10B104KB8NNNC	V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C503	100nF	CL10B104KB8NNNC	V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R502	10K	RC0603JR-0710KL	P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R501	10K	RC0603JR-0710KL	P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK

Alternates MPN2–4 on each part: same value/package, >= voltage, same dielectric

STM6822 supervisor, 3.08 V threshold, 1.6 s watchdog, open-drain $\overline{\text{RST}}$, manual reset
Group 1 SOT-23-5_open-drain output: R pull-up fitted.
Push button drives MR
Alternates (MPN2): TPS3828-33DBVR

Replicates block: WD and RST TPS3823

Chinmay

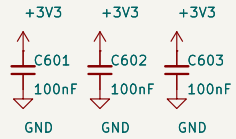
Sheet: /WD and RST STM6822/
File: WD_RST_STM6822.kicad_sch

Title: WD and RST STM6822

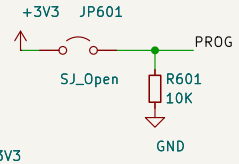
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 5/18

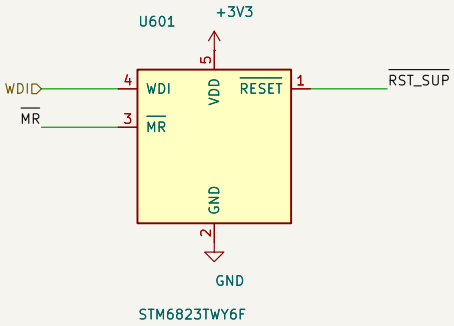
Watchdog and Reset controller — STM6823TWY6F

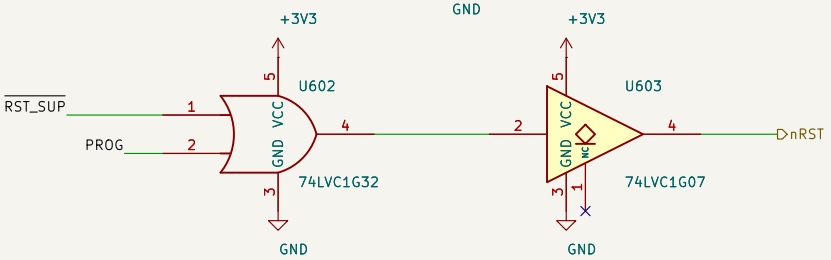
Decoupling: one 100nF per IC

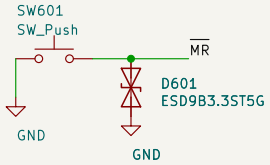


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C601 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C602 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C603 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R601 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

STM6823 supervisor, 3.08 V threshold, 1.6 s watchdog, push-pull $\overline{\text{RST}}$, manual reset
Group 1 SOT-23-5 (MAX823 pinout). Drop-in: TPS3823, MCP1316.
Push button drives MR
Alternates (MPN2): TPS3823-33DBVR, MCP1316T-29LE/OT

Replicates block: WD and RST TPS3823

Chinmay

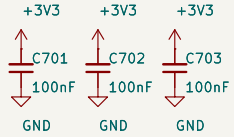
Sheet: /WD and RST STM6823/
File: WD_RST_STM6823.kicad_sch

Title: WD and RST STM6823

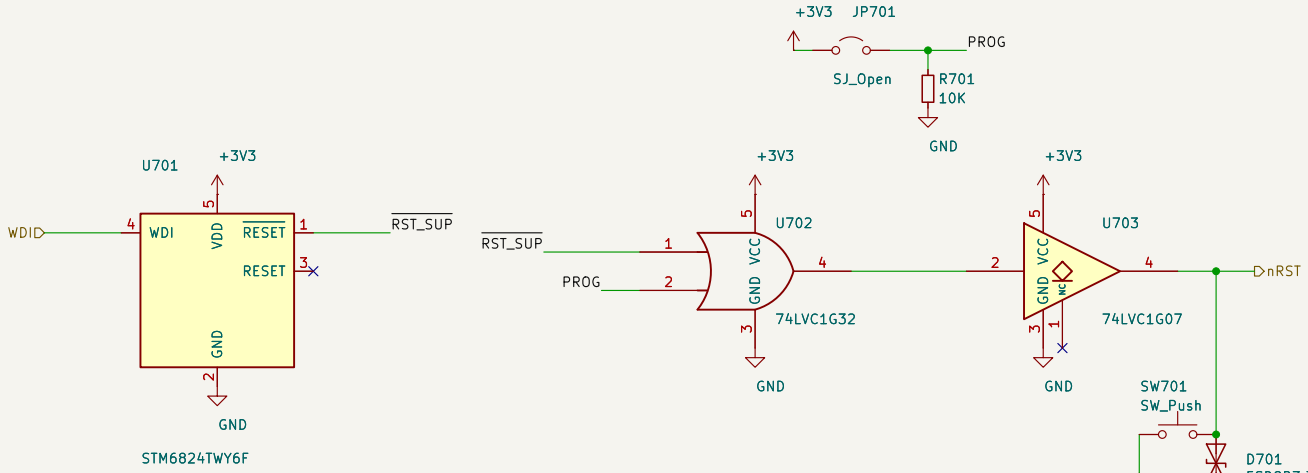
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 6/18

Watchdog and Reset controller — STM6824TWY6F

Decoupling: one 100nF per IC



Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash



PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C701 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C702 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C703 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R701 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

STM6824 supervisor, 3.08 V threshold, 1.6 s watchdog, push-pull $\overline{\text{RST}}$ + RST, no MR
Group 2 SOT-23-5 (MAX824 pinout): no MR pin, button resets nRST directly.
No MR pin on this IC: push button pulls nRST low directly
Alternates (MPN2): TPS3824-33DBVR, MCP1318T-29LE/OT

Replicates block: WD and RST TPS3823

Chinmay

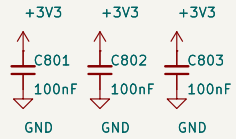
Sheet: /WD and RST STM6824/
File: WD_RST_STM6824.kicad_sch

Title: WD and RST STM6824

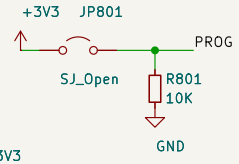
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 7/18

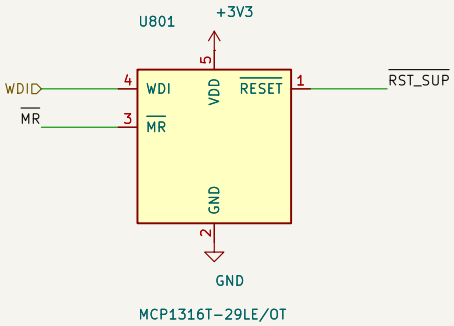
Watchdog and Reset controller – MCP1316T-29LE/OT

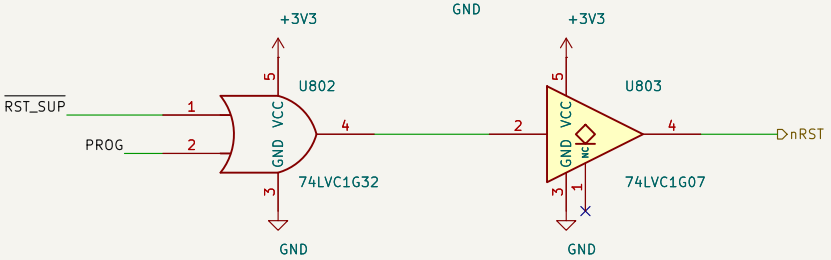
Decoupling: one 100nF per IC

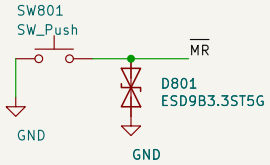


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C801 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C802 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C803 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R801 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

MCP1316 supervisor, 2.9 V threshold, watchdog, push-pull $\overline{\text{RST}}$, manual reset
Group 1 SOT-23-5. Check reset-delay / WDT codes in the order number.
Push button drives MR
Alternates (MPN2): TPS3823-33DBVR, STM6823TWY6F

Replicates block: WD and RST TPS3823

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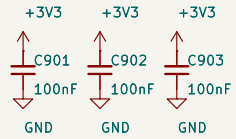
Sheet: /WD and RST MCP1316/
File: WD_RST_MCP1316.kicad_sch

Title: WD and RST MCP1316

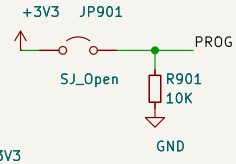
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 8/18

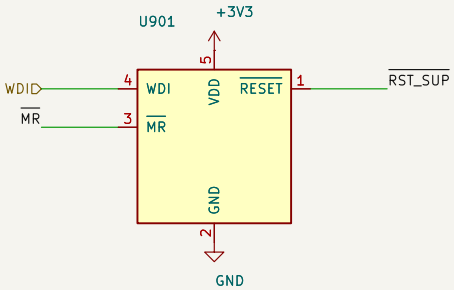
Watchdog and Reset controller – MCP1316MT-29LE/OT

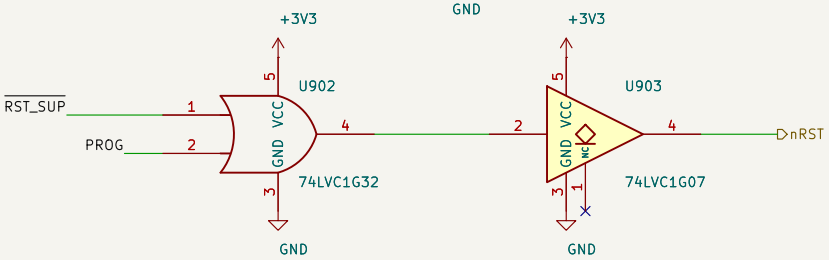
Decoupling: one 100nF per IC

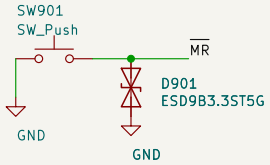


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









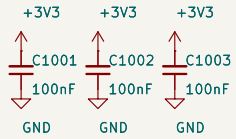
PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C901 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C902 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C903 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R901 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

MCP1316M supervisor, 2.9 V threshold, watchdog, open-drain $\overline{\text{RST}}$ with internal pull-up, manual reset
Group 1 SOT-23-5. Internal pull-up on RST: no external resistor.
Push button drives MR
Alternates (MPN2): –

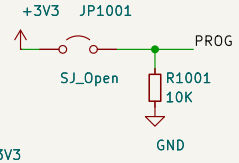
Replicates block: WD and RST TPS3823
Chinmay
Sheet: /WD and RST MCP1316M/
File: WD_RST_MCP1316M.kicad_sch
Title: WD and RST MCP1316M
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.5 | Id: 9/18

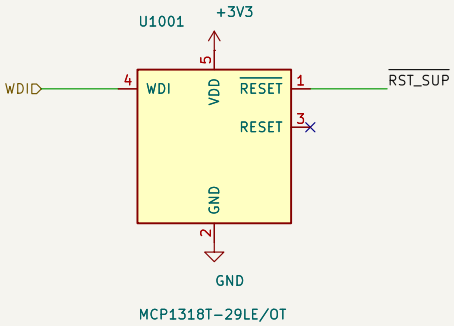
Watchdog and Reset controller – MCP1318T-29LE/OT

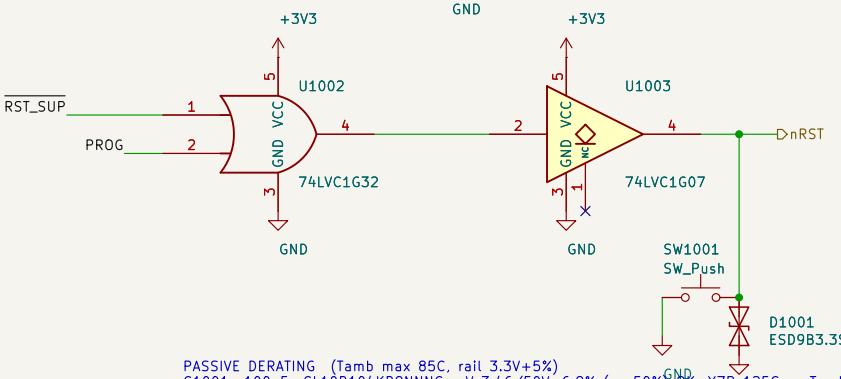
Decoupling: one 100nF per IC



Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash







PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1001 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1002 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1003 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1001 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

MCP1318 supervisor, 2.9 V threshold, watchdog, push-pull $\overline{\text{RST}}$ + RST, no MR
Group 2 SOT-23-5: no MR pin, button resets nRST directly.
No MR pin on this IC: push button pulls nRST low directly
Alternates (MPN2): TPS3824-33DBVR, STM6824TWY6F

Replicates block: WD and RST TPS3823

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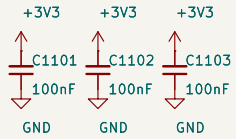
Sheet: /WD and RST MCP1318/
File: WD_RST_MCP1318.kicad_sch

Title: WD and RST MCP1318

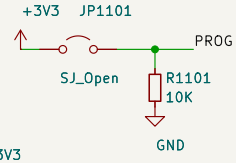
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 10/18

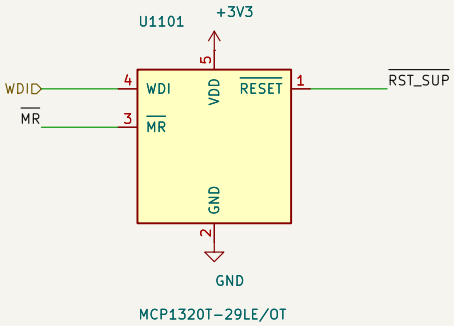
Watchdog and Reset controller – MCP1320T-29LE/OT

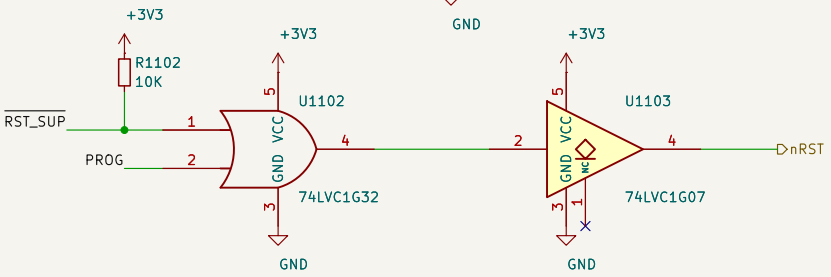
Decoupling: one 100nF per IC

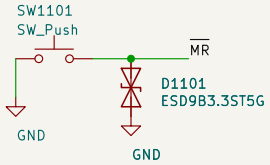


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1101 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1102 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1103 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1102 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1101 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

MCP1320 supervisor, 2.9 V threshold, watchdog, open-drain $\overline{\text{RST}}$ (external pull-up), manual reset
Group 1 SOT-23-5_open-drain output: R pull-up fitted.
Push button drives MR
Alternates (MPN2): TPS3828-33DBVR

Replicates block: WD and RST TPS3823

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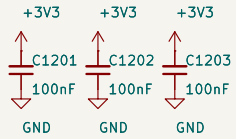
Sheet: /WD and RST MCP1320/
File: WD_RST_MCP1320.kicad_sch

Title: WD and RST MCP1320

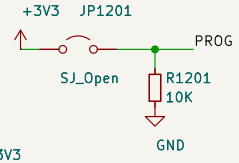
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 11/18

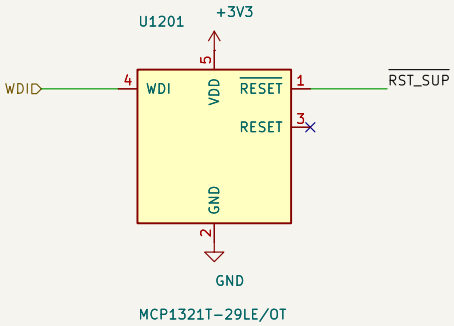
Watchdog and Reset controller – MCP1321T-29LE/OT

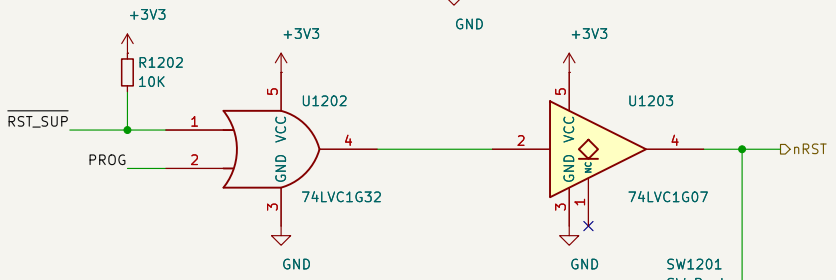
Decoupling: one 100nF per IC



Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash







PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1201 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1202 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1203 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1202 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1201 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2-4 on each part: same value/package, >= voltage, same dielectric

MCP1321 supervisor, 2.9 V threshold, watchdog, open-drain $\overline{\text{RST}}$ + push-pull RST, no MR
Group 2 SOT-23-5, open-drain RST: R pull-up fitted; button resets nRST directly.
No MR pin on this IC: push button pulls nRST low directly
Alternates (MPN2): –

Replicates block: WD and RST TPS3823

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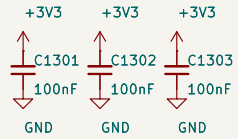
Sheet: /WD and RST MCP1321/
File: WD_RST_MCP1321.kicad_sch

Title: WD and RST MCP1321

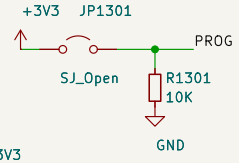
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 12/18

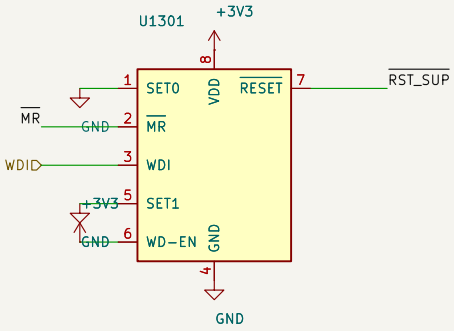
Watchdog and Reset controller — TPS35CA40KAGDDFR

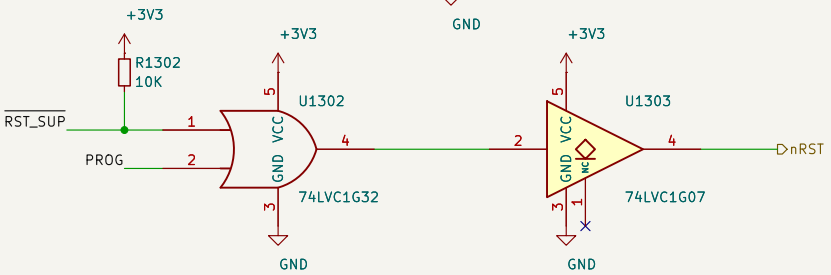
Decoupling: one 100nF per IC

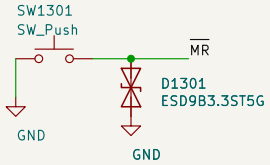


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









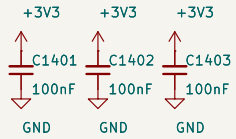
PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1301 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1302 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1303 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1302 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1301 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2–4 on each part: same value/package, >= voltage, same dielectric

TPS35 pinout C, 3.00 V threshold, 1.6 s watchdog, 200 ms reset, open-drain $\overline{\text{RESET}}$, 250 nA
Pinout option C (fixed timing). Order code built from the datasheet decoder: confirm it is released.
Push button drives MR
Alternates (MPN2): —

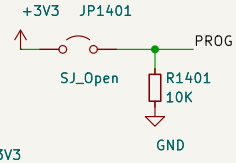
Replicates block: WD and RST TPS3823
Chinmay
Sheet: /WD and RST TPS35/
File: WD_RST_TPS35.kicad_sch
Title: WD and RST TPS35
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.5 | Id: 13/18

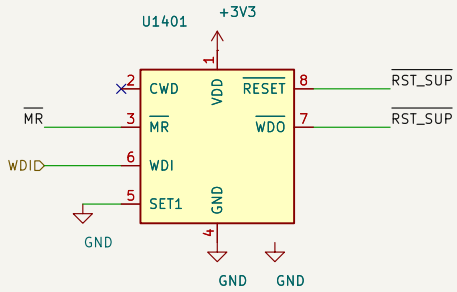
Watchdog and Reset controller – TPS3851G33SDRBR

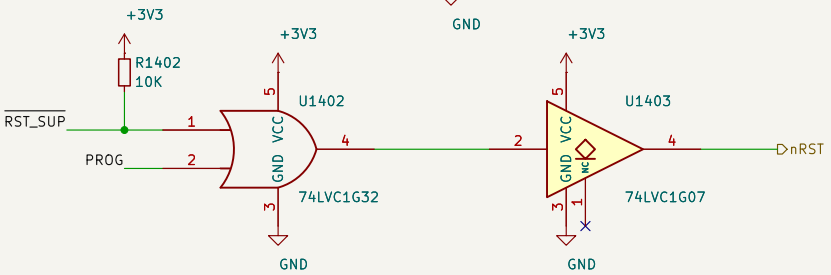
Decoupling: one 100nF per IC

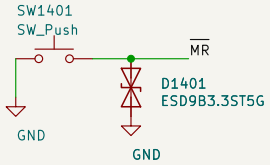


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1401 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1402 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1403 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1402 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1401 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2–4 on each part: same value/package, >= voltage, same dielectric

TPS3851 precision supervisor (0.8%), 3.3 V –4% threshold, watchdog, open-drain RESET + WDO
RESET and WDO wired-OR. CWD open = factory timing; check SET1 setting in datasheet.
Push button drives MR
Alternates (MPN2): –

Replicates block: WD and RST TPS3823

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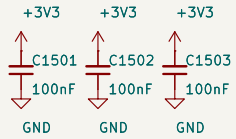
Sheet: /WD and RST TPS3851/
File: WD_RST_TPS3851.kicad_sch

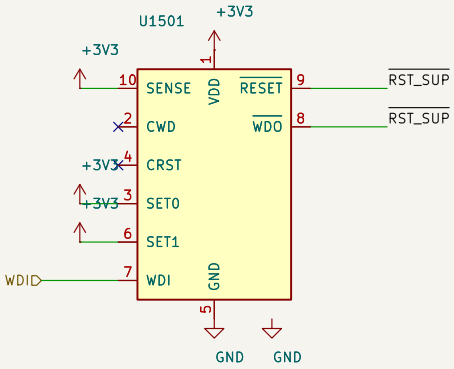
Title: WD and RST TPS3851

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 14/18

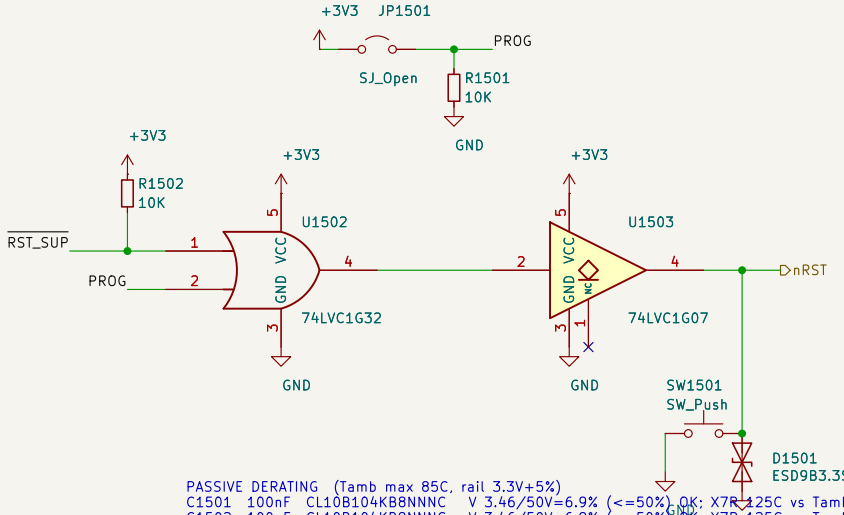
Watchdog and Reset controller – TPS3850G33DRCR

Decoupling: one 100nF per IC





Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash



PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1501 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1502 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1503 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1502 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1501 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2–4 on each part: same value/package, >= voltage, same dielectric

TPS3850 window-watchdog supervisor, 3.3 V −4%, SET0=SET1=1: 800–1600 ms window, open-drain outputs
WINDOW watchdog: WDI edges must arrive between 800 ms and 1600 ms. No MR: button resets nRST.
No MR pin on this IC: push button pulls nRST low directly
Alternates (MPN2): –

Replicates block: WD and RST TPS3823

Chinmay

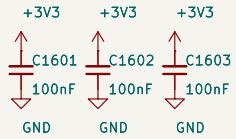
Sheet: /WD and RST TPS3850/
File: WD_RST_TPS3850.kicad_sch

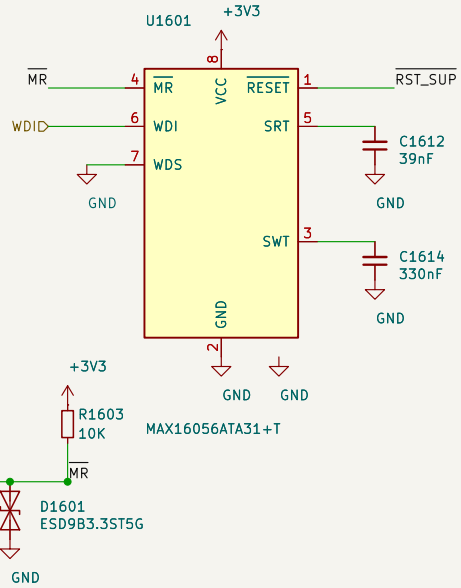
Title: WD and RST TPS3850

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 15/18

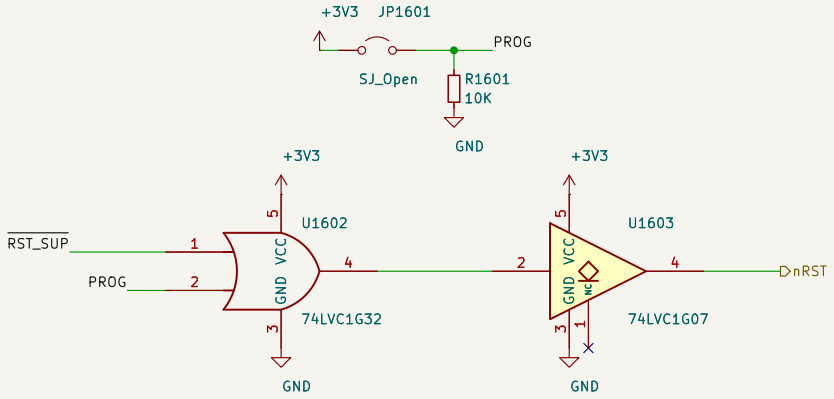
Watchdog and Reset controller — MAX16056ATA31+T

Decoupling: one 100nF per IC





Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash



PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1612 39nF CL10B393KB8NNNC V 1.30/50V=2.6% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1614 330nF CL10B334KO8NNNC V 1.30/16V=8.1% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1601 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1602 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1603 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1601 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1603 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2–4 on each part: same value/package, >= voltage, same dielectric

MAX16056 125 nA supervisor, 3.075 V threshold, 39 nF = 200 ms reset, 330 nF = 1.7 s watchdog
MR has no internal pull-up: R fitted. Footprint: check against Maxim land pattern 90-0059.
Push button drives MR
Alternates (MPN2): –

Replicates block: WD and RST TPS3823

Chinmay

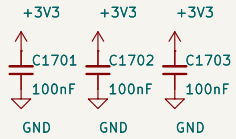
Sheet: /WD and RST MAX16056/
File: WD_RST_MAX16056.kicad_sch

Title: WD and RST MAX16056

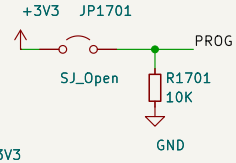
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 16/18

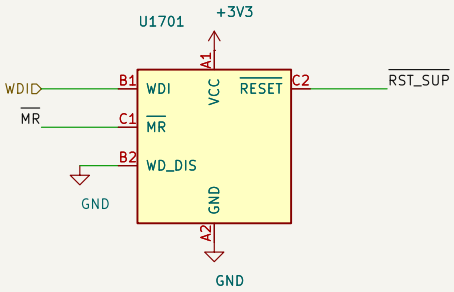
Watchdog and Reset controller – ADM8613Y232ACBZ-R7

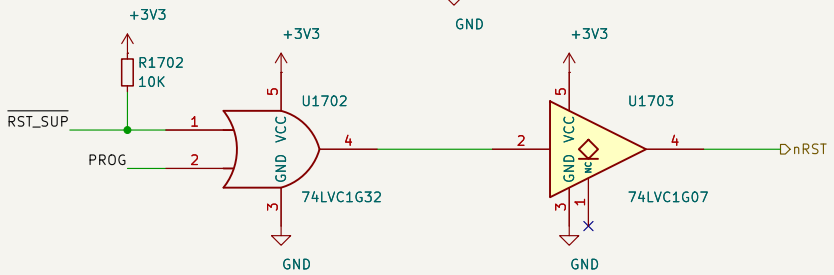
Decoupling: one 100nF per IC

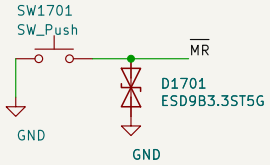


Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash









PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1701 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1702 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1703 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1702 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1701 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2–4 on each part: same value/package, >= voltage, same dielectric

ADM8613 92 nA supervisor, WLCSP-6, 1.6 s watchdog, open-drain $\overline{\text{RESET}}$
Standard part has a 2.32 V threshold; 2.93/3.08 V are non-standard orders.
Push button drives MR
Alternates (MPN2): –

Replicates block: WD and RST TPS3823

Chinmay

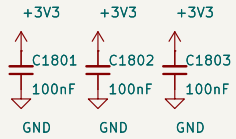
Sheet: /WD and RST ADM8613/
File: WD_RST_ADM8613.kicad_sch

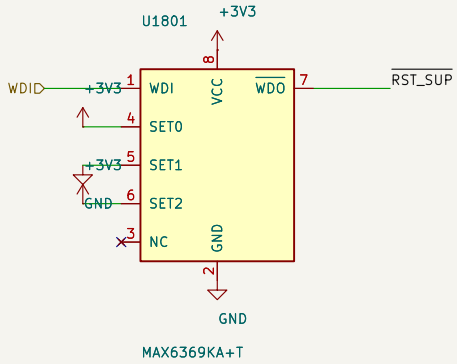
Title: WD and RST ADM8613

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 17/18

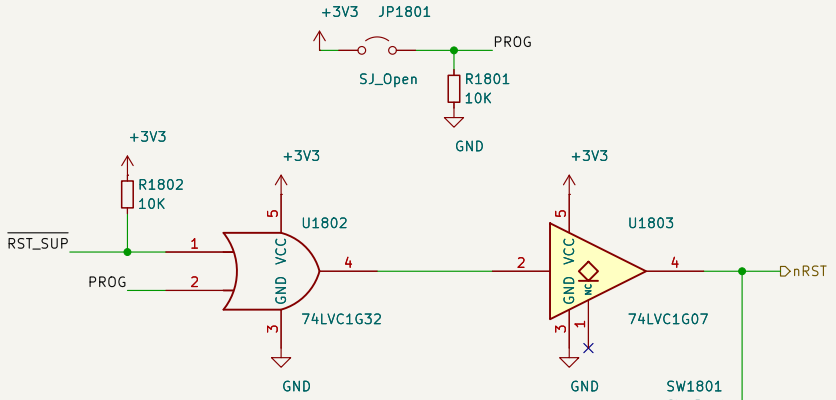
Watchdog and Reset controller – MAX6369KA+T

Decoupling: one 100nF per IC





Bridge jumper JP during flash
This circuit avoids the WD controller resetting the controller everytime you try to flash



PASSIVE DERATING (Tamb max 85C, rail 3.3V+5%)
C1801 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1802 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
C1803 100nF CL10B104KB8NNNC V 3.46/50V=6.9% (<=50%) OK; X7R 125C vs Tamb 85C OK
R1802 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
R1801 10K RC0603JR-0710KL P 1.20mW/82.4mW@85C=1.5% (<=50%) OK; V 3.46/50V=6.9% (<=80%) OK
Alternates MPN2–4 on each part: same value/package, >= voltage, same dielectric

MAX6369 pin-selectable watchdog (SET2..0 = 101: 1–3 s), open-drain $\overline{WDO}$, NO voltage monitor
Watchdog only: pair with a voltage supervisor. No MR: button resets nRST.
No MR pin on this IC: push button pulls nRST low directly
Alternates (MPN2): –

Replicates block: WD and RST TPS3823
Chinmay
Sheet: /WD and RST MAX6369/
File: WD_RST_MAX6369.kicad_sch
Title: WD and RST MAX6369
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.5 | Id: 18/18